

# Data warehouse data modeling

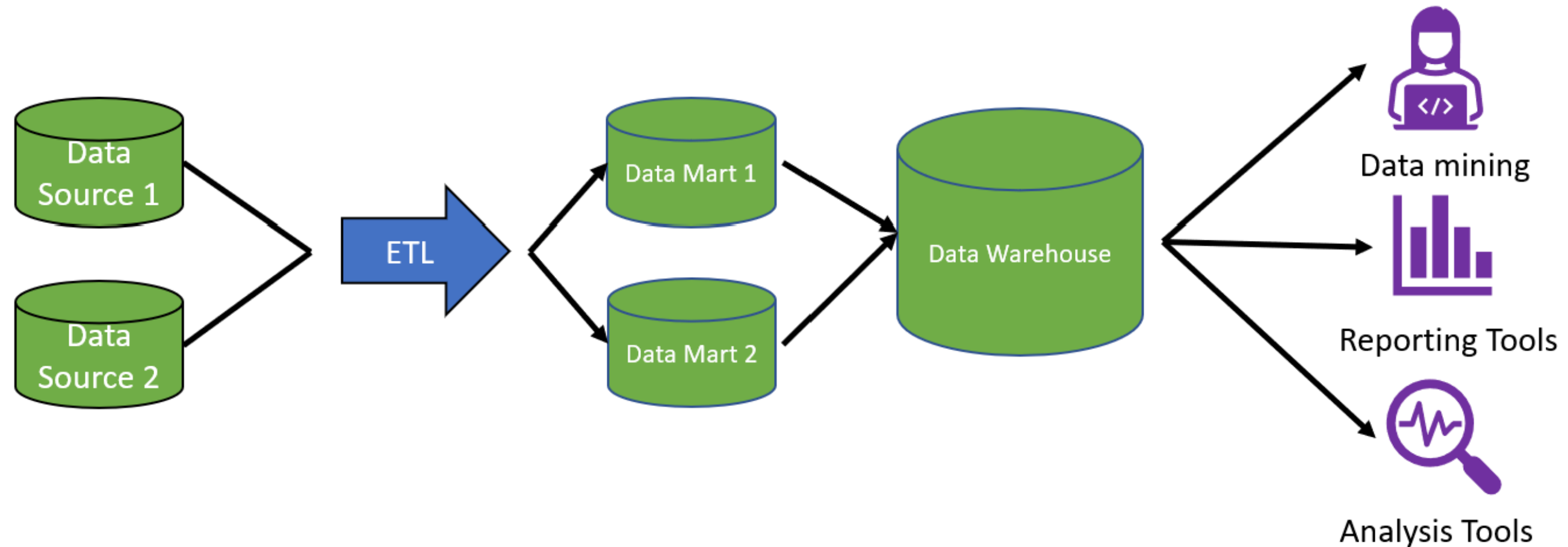
DATA WAREHOUSING CONCEPTS



**Aaren Stubberfield**  
Data Scientist

# Data models

- Bottom-up, Kimball model = star & snowflake schemas
- Denormalized data models



# It's Bravo again!

- Hypothetical publicly traded company
  - Sells home office furniture



<sup>1</sup> Photo from Pexels by Pixabay

# Fact table

- Measurements, metrics, or facts about an organization
- Links to dimension tables for more details

**Table Name:** Sales\_Order\_Fact

| Keys | ColumnName  |
|------|-------------|
| FK   | CustomerID  |
| FK   | DateID      |
| FK   | ProductID   |
|      | UnitSold    |
|      | SalesAmount |
|      | Tax         |

**Legend:** FK = Foreign Key

# Dimension table

- Dimensions/attributes about a process
- Holds reference data
- Dimension tables add more detail to fact table

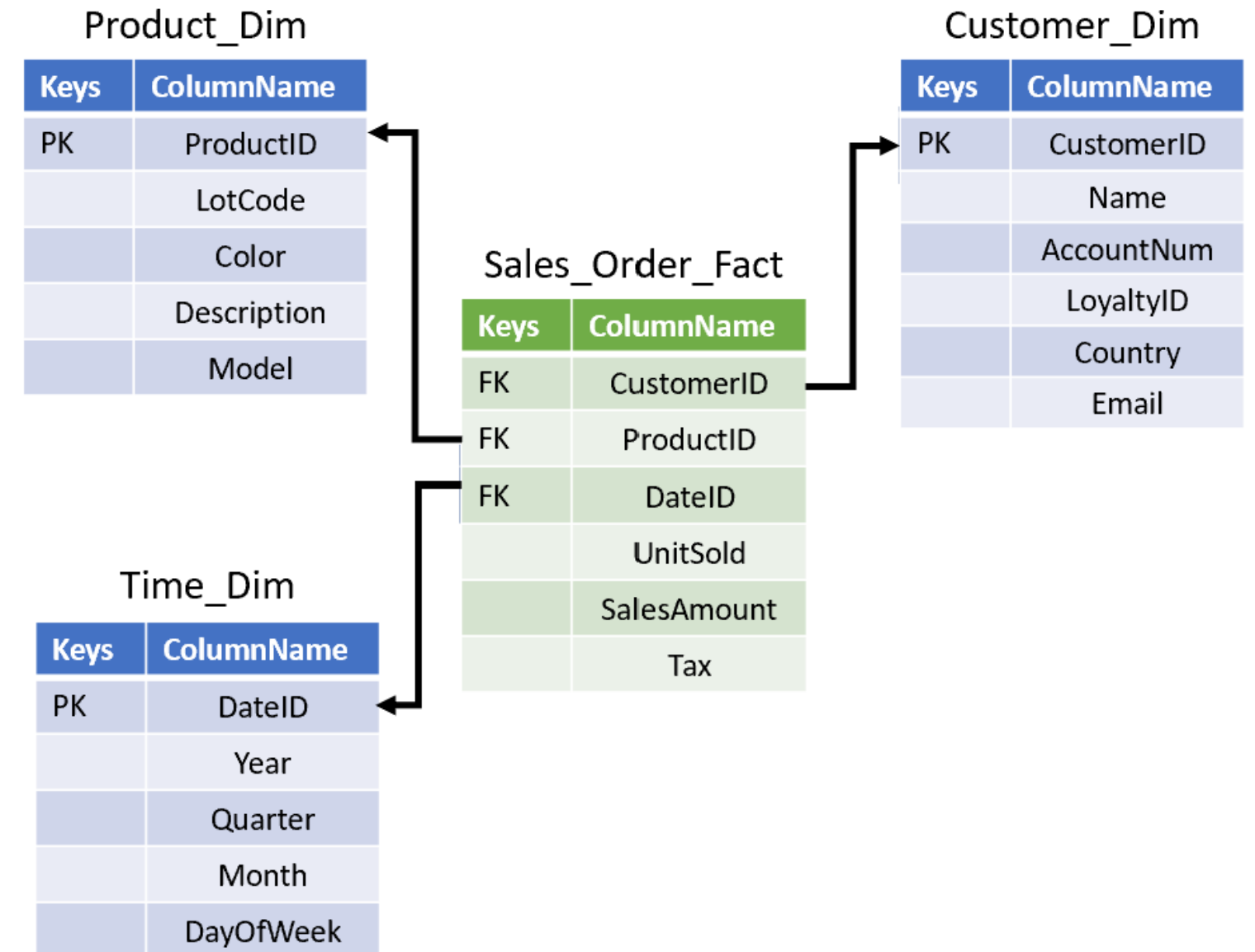
**Table Name:** Customer\_Dim

| Keys | ColumnName |
|------|------------|
| PK   | CustomerID |
|      | Name       |
|      | AccountNum |
|      | LoyaltyID  |
|      | Country    |
|      | Email      |

**Legend:** PK = Primary Key

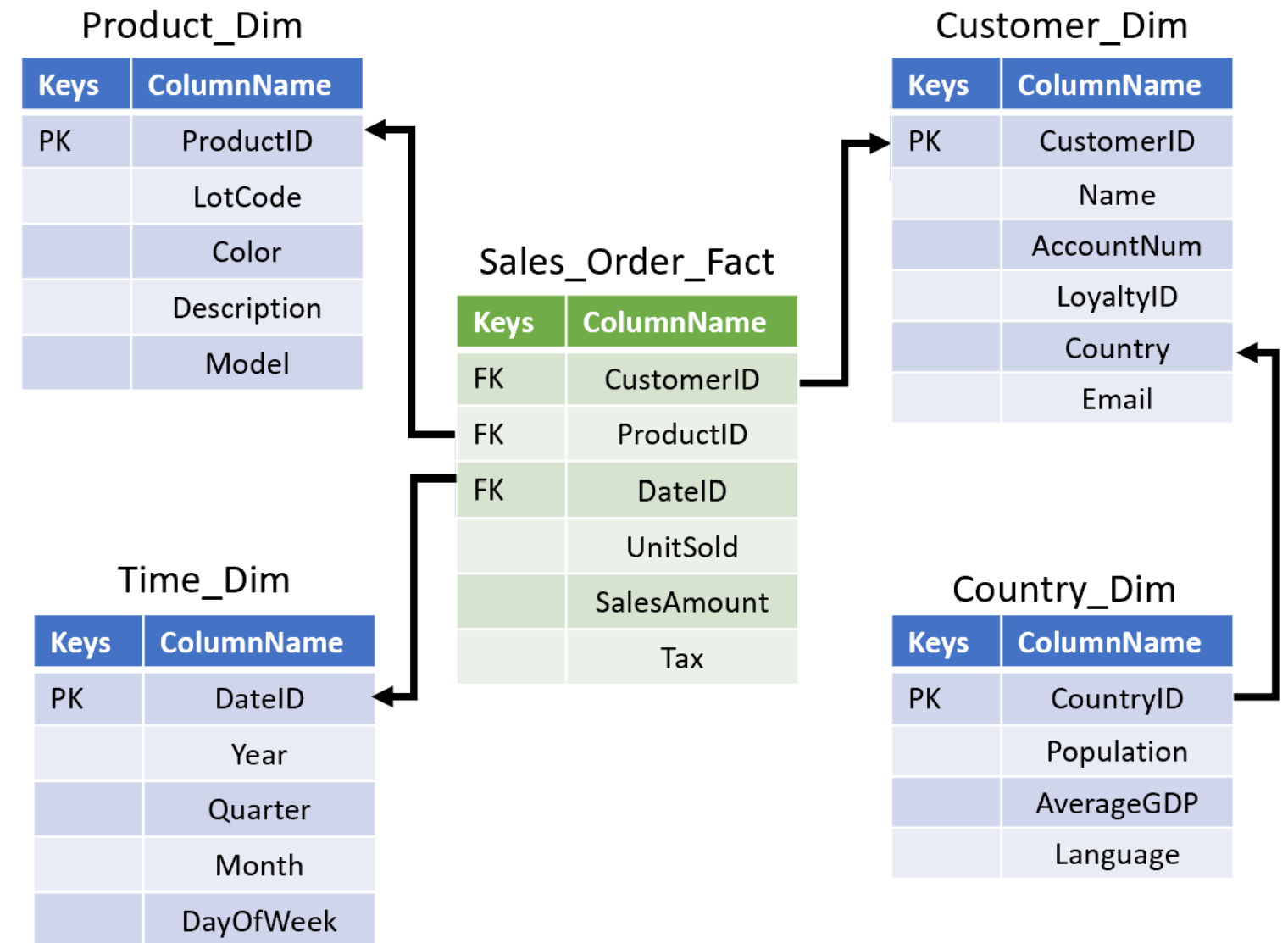
# Star schema

- A central fact table, with one or more dimensional tables
- Easy for business users



# Snowflake schema

- Dimensional table connected through another dimensional table





**Let's practice!**  
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# Kimball's four step process

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# Step 1 - Select the organizational process

- Ask questions about a process
- Kimball bottom-up approach starts with a business process



## Examples of organizational processes:

- Invoice and billing
- Product quality monitoring
- Marketing

# Step 2 - Declare the grain

- Grain = level to store fact table
- A level of data that cannot be split further

## Examples of business processes:

- Music service -> Song grain
- Shipping service -> Line item grain

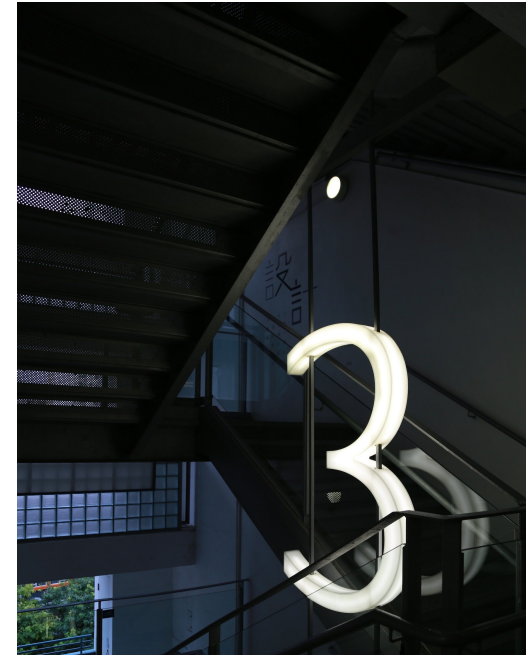


# Step 3 - Identify the dimensions

- Choose dimensions that apply to each row
- How to describe the data?
- Business users and analysts = valuable feedback

## Examples of common dimensions:

- **Time:** year, quarter, and month
- **Location:** address, state, and country
- **Users:** names and email address



<sup>1</sup> Photo by Alison Pang on Unsplash

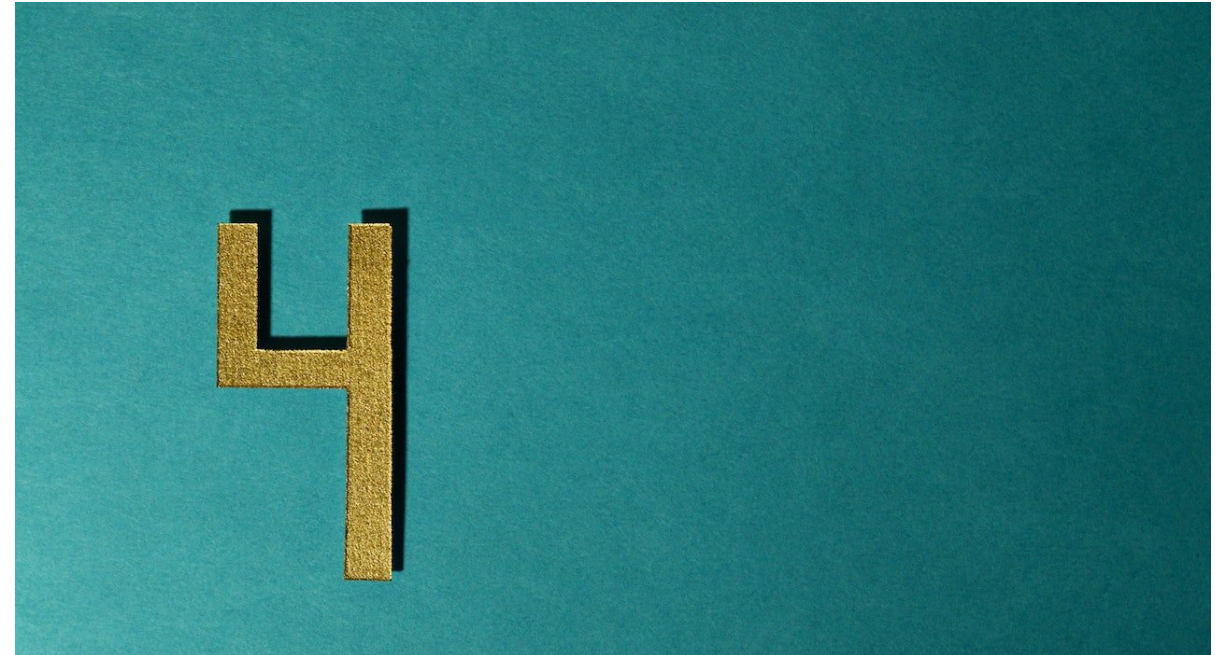


# Step 4 - Identify the facts

- Numerical facts for each fact table row
- What are we answering?
- Metrics should be true at selected grain

## Examples of facts:

- Music service: total number of plays, sales revenue of a song
- Ride-sharing: travel distance, time needed



<sup>1</sup> Photo by Miguel Á. Padriñán

# Summary

## Steps:

1. Select the organizational process.
2. Declare the grain.
3. Identify the dimensions.
4. Identify the facts.

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# Slowly changing dimensions

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# The challenge

Original

| ProductID | Description  | Category      |
|-----------|--------------|---------------|
| 12345     | Tesla-ModelY | electric-veh. |



Update Category:

- **Current:** electric-veh.
- **New:** electric-crossover

# Type I

- Update value in table
- Will lose any history

## Original

| ProductID | Description  | Category                 |
|-----------|--------------|--------------------------|
| 12345     | Tesla-ModelY | <del>electric-veh.</del> |

## New

| ProductID | Description  | Category           |
|-----------|--------------|--------------------|
| 12345     | Tesla-ModelY | electric-crossover |

# Type II

- Add a row with the updated value
- The history is retained

## Original

| ProductID | Description  | Category      |
|-----------|--------------|---------------|
| 12345     | Tesla-ModelY | electric-veh. |

## New

| ProductID | Description  | Category           | StartDate  | EndDate    |
|-----------|--------------|--------------------|------------|------------|
| 12345     | Tesla-ModelY | electric-veh.      | 1970-01-01 | 2022-03-10 |
| 20053     | Tesla-ModelY | electric-crossover | 2022-03-11 | 2050-12-31 |

# Type III

- Add column to dimension table to track changes
- Can view past and current data together
- Can require reporting changes and limited tracking

## Original

| ProductID | Description  | Category      |
|-----------|--------------|---------------|
| 12345     | Tesla-ModelY | electric-veh. |

## New

| ProductID | Description  | Category           | PastCategory  |
|-----------|--------------|--------------------|---------------|
| 12345     | Tesla-ModelY | electric-crossover | electric-veh. |

# Modern approach

- Snapshot the whole dimension table
- Use historical snapshots for historical reports

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# Row vs. column data store

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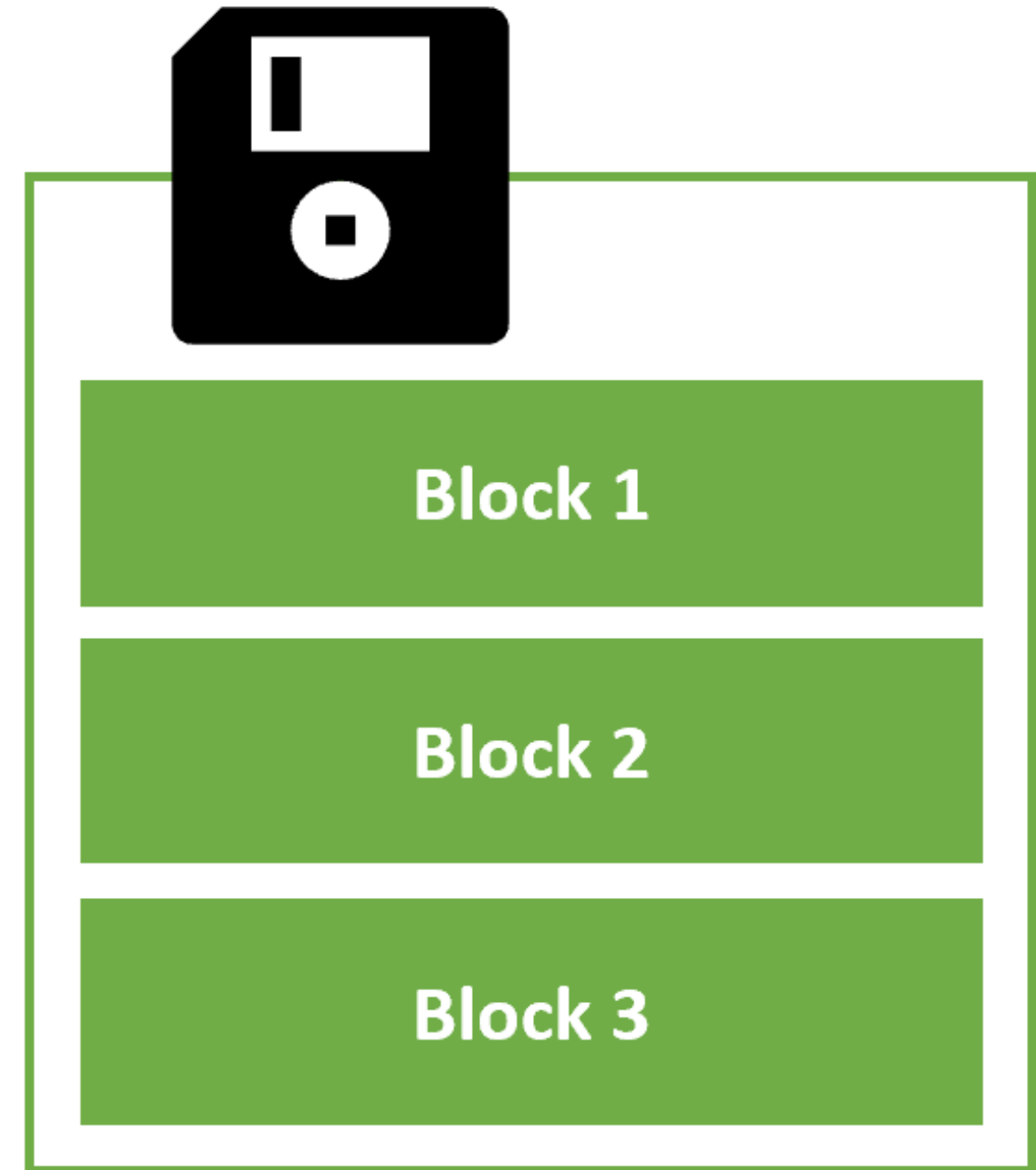
# Why is it important?

- Optimizing queries for speed
- Column store format for data warehouse tables is best for analytic workloads



# Basics of computer storage

- Computers store data in blocks.
- Reads the required blocks when retrieving data.
- Reading fewer blocks increases the overall speed of the process.



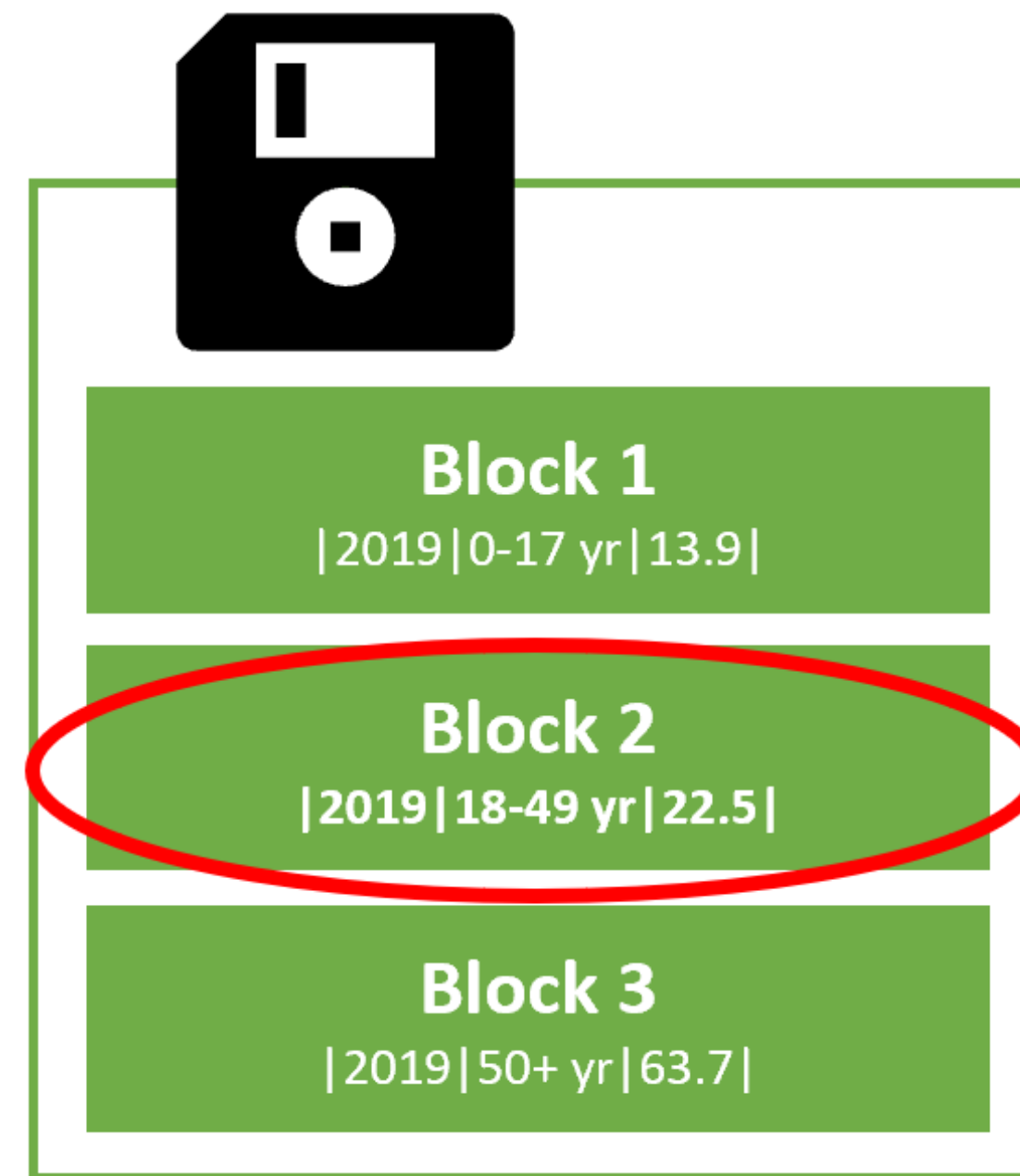
# Example of health table

- CDC (Centers for Disease Control and Prevention)
- Flu infection data by age groups over multiple seasons

| SEASON | AGE GROUP | HOSPITALIZATION PERCENTAGE |
|--------|-----------|----------------------------|
| 2019   | 0-17 yr   | 13.9%                      |
| 2019   | 18-49 yr  | 22.5%                      |
| 2019   | 50+ yr    | 63.7%                      |
| 2020   | 0-17 yr   | 3.9%                       |
| 2020   | 18-49 yr  | 18.1%                      |
| 2020   | 50+ yr    | 78%                        |
| 2021   | 0-17 yr   | 15.6%                      |
| 2021   | 18-49 yr  | 23.3%                      |
| 2021   | 50+ yr    | 61.1%                      |

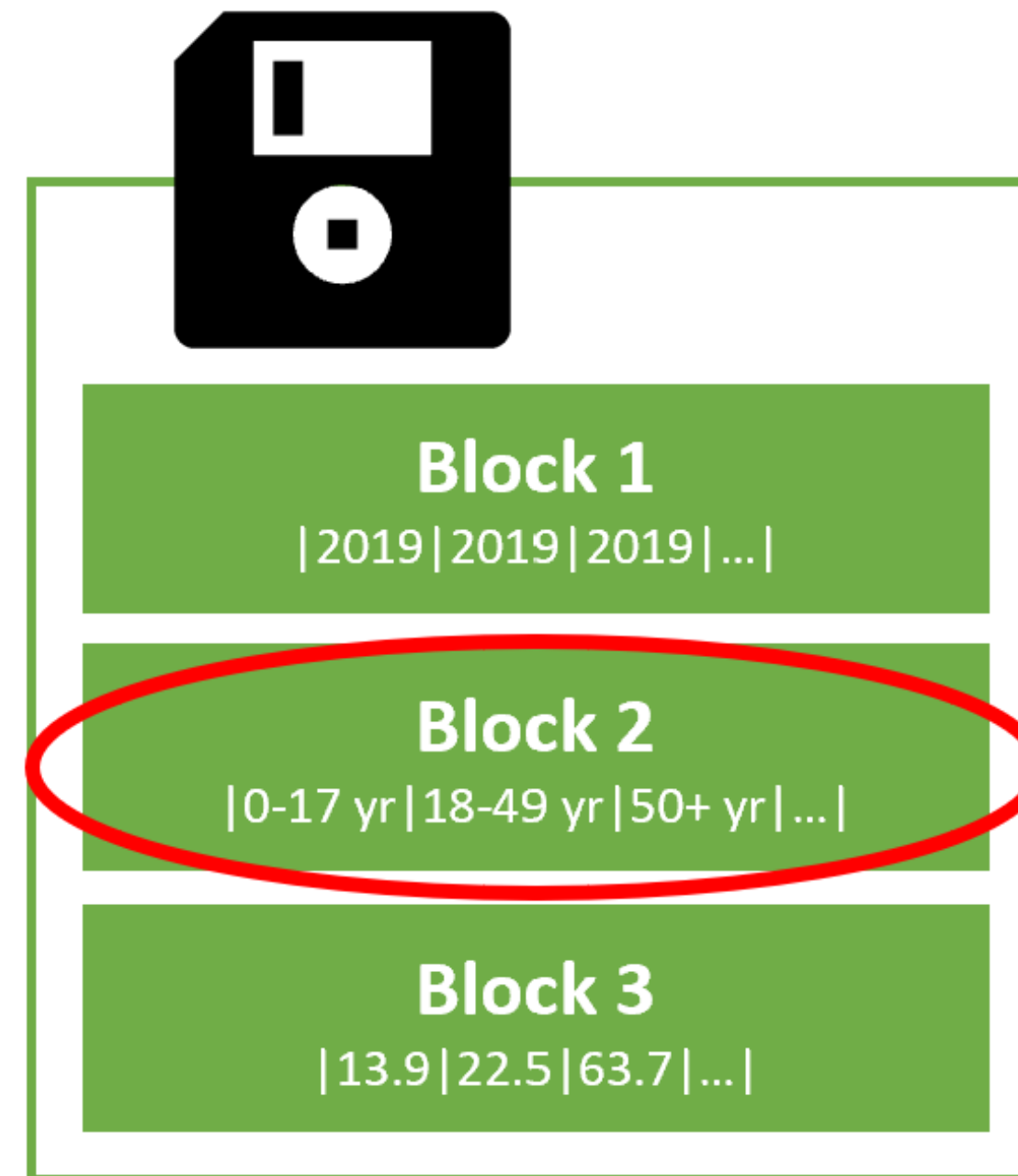
# Row store example

| SEASON      | AGE GROUP       | HOSPITALIZATION PERCENTAGE |
|-------------|-----------------|----------------------------|
| 2019        | 0-17 yr         | 13.9%                      |
| <b>2019</b> | <b>18-49 yr</b> | <b>22.5%</b>               |
| 2019        | 50+ yr          | 63.7%                      |
| 2020        | 0-17 yr         | 3.9%                       |
| 2020        | 18-49 yr        | 18.1%                      |
| 2020        | 50+ yr          | 78%                        |
| 2021        | 0-17 yr         | 15.6%                      |
| 2021        | 18-49 yr        | 23.3%                      |
| 2021        | 50+ yr          | 61.1%                      |



# Column store example

| SEASON | AGE GROUP       | HOSPITALIZATION PERCENTAGE |
|--------|-----------------|----------------------------|
| 2019   | <i>0-17 yr</i>  | 13.9%                      |
| 2019   | <i>18-49 yr</i> | 22.5%                      |
| 2019   | <i>50+ yr</i>   | 63.7%                      |
| 2020   | <i>0-17 yr</i>  | 3.9%                       |
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| 2021   | <i>50+ yr</i>   | 61.1%                      |



# Summary

## Row Store

- Row data is stored together in blocks
- Ideal for transactional workloads

## Column Store

- Column data is stored together in blocks
- Ideal for analytical workloads
- Better data compression

# It's practice time!

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